

MARKET SENTIMENT
x TRADER PERFORMANCE

ANALYSIS REPORT

Uncovering Hidden Patterns in Bitcoin Fear & Greed vs Hyperliquid Trading Data

50,000	200	731	8
Total Trades Analysed	Unique Traders	Days of Sentiment Data	Symbols Covered

PrimeTrade.ai — Data Science Assignment Submission — 2024

01 EXECUTIVE SUMMARY

This analysis explores the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance on the Hyperliquid decentralised exchange. Across 731 days of daily sentiment data merged with 50,000 trade records from 200 unique accounts, six actionable patterns emerge that can directly inform smarter trading strategies.

Dataset Overview

Dataset	Key Columns	Records
Fear & Greed Index	date, value (0-100), classification (5 categories)	731 days
Hyperliquid Trader Data	account, symbol, execution_price, size, side, time, start_position, event, closedPnL, leverage, notional_value	50,000 trades

Sentiment Distribution

The 731-day window (Jan 2023 – Dec 2024) was dominated by Extreme Fear (37.5%), reflecting the volatile crypto environment. Greed and Extreme Greed together account for ~31.6% of days, providing meaningful contrast for comparative analysis. Neutral phases (9.7%) are the rarest.

Methodology

Each trade record was joined to the Fear & Greed dataset on calendar date. Sentiment phases were classified into five buckets per the standard index thresholds: Extreme Fear (0-24), Fear (25-44), Neutral (45-55), Greed (56-75), Extreme Greed (76-100). Statistical significance was assessed via Pearson correlation, ANOVA, and rolling-window analysis.

02 OVERVIEW DASHBOARD

The six-panel overview dashboard below presents distribution, volume, profitability, win rate, leverage habits, and asset preference sliced by all five sentiment categories, together with a full monthly PnL vs Fear & Greed time series.



Key Observations from the Overview

- Win Rate:** Neutral-phase trading shows the highest win rate (~52%), followed by Fear. Extreme Greed has the lowest win rate, suggesting crowded, late-entry positions.
- Volume:** Trade volume is highest in Extreme Fear — indicating that opportunistic traders are most active when others panic.
- Leverage:** Average leverage peaks during Greed phases (+15% vs Fear baseline), increasing liquidation risk at precisely the wrong moment.
- Symbol Preference:** BTC and ETH maintain stable share across all sentiment phases; speculative tokens (DOGE, MATIC) spike in Greed phases.

03 TRADER BEHAVIOUR DEEP DIVE

This section dissects individual trader behaviour patterns: PnL distributions, liquidation rates, buy/sell directionality, top vs bottom performer trajectories, and the leverage-PnL scatter.



Key Observations from Behaviour Analysis

PnL Distributions: Extreme sentiment phases (both Fear and Greed) exhibit fat tails on the loss side, pointing to higher variance and increased blow-up risk at the extremes.

Liquidation Rate: Liquidations cluster in Extreme Greed phases — leverage-chasing during bull runs leads to cascading wipeouts when price reverses.

Buy/Sell Imbalance: BUY orders dominate in Greed phases (~58% BUY), while SELL pressure increases during Fear — consistent with retail FOMO/panic behaviour.

Top vs Bottom Traders: The top 5 traders by cumulative PnL show steady upward equity curves; the bottom 5 suffer from deep drawdowns concentrated in Greed phases.

04 STATISTICAL ANALYSIS

Correlation analysis, event-type PnL breakdown, rolling sentiment-performance correlation, and trader profiling complete the quantitative picture.

Statistical Analysis & Strategy Insights



KEY INSIGHTS & STRATEGY RECOMMENDATIONS

- CONTRARIAN EDGE IN EXTREME FEAR** -> Trades placed during Extreme Fear show highest avg PnL, validating the buy-when-fearful principle.
- LEVERAGE DISCIPLINE** -> Unprofitable traders use 1.3x more leverage on average. Capping leverage at 5x during extremes improves risk-adjusted returns.
- LIQUIDATION CLUSTERING** -> Liquidation events spike +40% during Extreme Greed phases. Crowded longs unwind fast. Reduce exposure when FGI > 75.
- TAKE-PROFIT DISCIPLINE** -> TAKE_PROFIT events yield 2.1x higher avg PnL than ORDER_FILLED. Pre-set exits outperform discretionary closes.
- SYMBOL ROTATION** -> BTC & ETH dominate in Fear phases; SOL/DOGE/AVAX see higher activity in Greed phases. Rotate accordingly.
- ROLLING CORRELATION SIGNAL** -> 30-day rolling r between FGI and daily PnL flips sign. Positive r = trend-following; Negative r = contrarian edge active.

Statistical Highlights

Leverage vs PnL correlation: $r = -0.18$ ($p < 0.001$) — higher leverage is weakly but significantly associated with worse PnL outcomes, especially in high-sentiment phases.

Event-Type PnL: TAKE_PROFIT events average 2.1x higher PnL than unplanned ORDER_FILLED exits. LIQUIDATION events are the single largest driver of negative PnL.

Rolling 30-Day Correlation: The sign of the rolling r (FGI vs Daily PnL) flips regularly, implying the relationship is regime-dependent — not a static signal.

Profitable vs Unprofitable Traders: Profitable traders show a 12% higher win rate, 22% lower average leverage, and more conservative position sizing.

05 KEY INSIGHTS & STRATEGY RECOMMENDATIONS

1	Contrarian Edge in Extreme Fear Trades executed during Extreme Fear phases generate the highest median PnL. The data validates the buy-when-fearful principle — but only with defined risk limits.	BUY THE DIP
2	Leverage Discipline is Critical Unprofitable traders average 1.3x more leverage. Capping leverage at 5x during sentiment extremes (FGI < 25 or > 75) meaningfully improves risk-adjusted returns.	RISK MGMT
3	Liquidation Clustering in Extreme Greed Liquidation rates jump +40% when FGI > 75. Crowded long positions unwind rapidly on even minor price corrections. Reduce gross exposure when FGI exceeds 75.	REDUCE RISK
4	Pre-Set Exits Outperform Discretionary Closes TAKE_PROFIT events produce 2.1x better average PnL vs discretionary ORDER_FILLED exits. Always define exit levels before entering a position.	SET TARGETS
5	Rotate Symbols with Sentiment BTC and ETH retain share in all phases (safe-haven crypto). Rotate into higher-beta tokens (SOL, AVAX, DOGE) only in confirmed Greed phases with tight stops.	ROTATE
6	Monitor Rolling FGI-PnL Correlation The 30-day rolling correlation between the FGI and daily PnL flips sign regularly. When $r > 0$, trend-following strategies work; when $r < 0$, lean contrarian.	REGIME WATCH

06 CONCLUSIONS & FUTURE WORK

Summary

The Fear & Greed Index is a useful but non-linear signal. Its relationship with trader PnL is regime-dependent: during Extreme Fear it predicts contrarian profitability; during Extreme Greed it predicts liquidation risk. The most profitable traders in the dataset systematically use lower leverage, pre-set exits, and show greater asset diversification across sentiment phases.

Actionable Trading Framework

FGI Range	Sentiment	Recommended Action	Max Leverage
0 – 24	Extreme Fear	Accumulate BTC/ETH; set take-profits	3x
25 – 44	Fear	Selective longs; tight stops	5x
45 – 55	Neutral	Balanced; follow momentum	5x
56 – 75	Greed	Rotate to beta assets; reduce size	3x
76 – 100	Extreme Greed	Take profits; avoid new longs; hedge	1x

Future Work

- On-chain data integration (exchange inflows/outflows, whale wallet movements) to enrich sentiment signals beyond the Fear & Greed Index.
- Machine learning classifier (XGBoost / LSTM) trained on FGI + order-flow features to predict next-day win probability per sentiment regime.
- Real-time dashboard integrating live FGI API data with position-level risk alerts.
- Cross-exchange study to assess whether the same sentiment-performance patterns hold on centralised venues (Binance, Bybit) vs decentralised (Hyperliquid, dYdX).

Submitted for PrimeTrade.ai Data Science Role | Analysis period: Jan 2023 – Dec 2024 | Tools: Python, pandas, matplotlib, scipy, reportlab